

WEEDS & THEIR CLASSIFICATION

WEEDS

Weeds are plants that grow where they are not wanted, often competing with cultivated crops or native vegetation

Harmful Impacts Of Weed On Agriculture

↓ Competition with Crops:

Weeds **compete vigorously with cultivated crops** for essential resources such as water, nutrients, space and sunlight.

↓ Allelopathy:

Certain weeds, such as **black walnut (*Juglans nigra*)**, release chemicals into the **soil that inhibits the growth of neighboring plants**, a phenomenon known as allelopathy. This not only reduces crop productivity but also complicates weed management efforts

↓ Hosts for Pests and Diseases:

Weeds can serve as hosts for **pests and diseases that affect crops, providing a reservoir for pathogens to thrive and spread**. For instance, common *Chenopodium album* can harbor insect pests that then move onto nearby crops, causing widespread damage.

Ecological Consequences

↓ Reduction in Biodiversity:

Invasive weeds **often outcompete native plants**, leading to a decline in biodiversity within ecosystems. This loss of native species can disrupt ecological balance and diminish habitats for wildlife

↓ Habitat Alteration:

Weeds can alter **habitat structure and composition**, impacting wildlife populations and ecological processes.

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↓ Soil Degradation:

Weeds with shallow root systems or **invasive growth habits can contribute to soil erosion, especially in vulnerable landscapes**. This erosion leads to loss of fertile topsoil and degradation of soil quality.

Human Health And Safety

↓ Allergies and Irritation:

Weeds like **ragweed (Ambrosia spp.) produce abundant pollen** that triggers allergic reactions in susceptible individuals, causing respiratory issues and seasonal discomfort.

↓ Toxicity and Poisoning:

Some weeds contain **toxins that pose risks to human and animal health if ingested**. For instance, poison **Conium maculatum and Datura stramonium are poisonous plants** that can cause severe poisoning symptoms

Economic And Social Impacts

↓ Cost of Control:

Controlling weed infestations requires **significant financial resources in terms of herbicides, labor, and management practices**. Failure to adequately control weeds can lead to substantial economic losses for agricultural producers and land managers.

↓ Impacts on Recreation and Aesthetics:

Weeds can diminish the aesthetic value of landscapes, parks, and recreational areas. **Invasive aquatic weeds, such as water hyacinth (Eichhornia crassipes), can clog waterways**, impeding boating, fishing, and other recreational activities.

Spread And Invasive Potential

↓ Rapid Dispersal:

Weeds often **have mechanisms for rapid dispersal and colonization** of new habitats, facilitated by adaptations such as prolific seed production, wind dispersal, or attachment to animals and human activities

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↓ Invasive Traits:

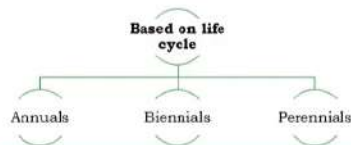
Some weeds possess traits that enable them to thrive in diverse environmental conditions and outcompete native vegetation

Common Weeds & Their Effect

Weeds Species	Harmful Effects
Bindweed	Competes with crops for nutrients, water and sunlight leads to reduce crop yields
Wild mustard	Harbors pests and diseases that can spread to nearby crops, increasing crop damage
Ragweed	Harbors pests and diseases than can affect crop health.
Black walnut	Release chemicals(allelochemicals) that inhibit the growth of nearby plants, including crops and this process known as allelopathy
Jhonson grass	Competes with crops for nutrients water and spread very quickly.
Morning glory	Interferes with farm equipment and structures, covering and damaging them
Dandelions	Reduces aesthetic value in gardens, landscapes, requiring constant removal.
Crab grass	Crowding out desired plants.
Nightshade	Toxic to livestock if ingested, potentially leading to illness or death.
Milkweed	Toxic to livestock if ingested, causing digestive issues and other health problems.
Water hyacinth	Invasive species that forms dense mats, impacts water flow and ecosystem imbalance.
Broom	Increase fire risk and displaces native vegetation, disrupting ecosystems

Classification Of Weeds

There are more than 3,50,000 plant species around the world, of which 30,000 have been recognized as weed species

1. Based on Life cycle

Weed classes based on **life cycle/span** aren't continuously steady since the developing term of a few weeds' changes beneath the impact of climatic components.

1. Annuals

Annual weeds usually **germinate, grow and produce seeds** within a season/year and die up.

It may be of **four types according to the season** they grow viz

1. Summer season annuals:

- ✓ They germinate during the summer season if there is ample moisture in soil, otherwise, late in the season when there are splashes of rainfall
- ✓ e.g. *Ageratum conyzoides*, *Digera arvensis*, *Digitaria* spp, *Acrachne racemosa*, *Dactyloctenium aegyptium*, are summer season annuals



Digera Arvensis

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4. **Rainy/wet season annuals (Kharif annuals)**

- ✓ They germinate **during the rainy/wet season when there is ample moisture** in soil conserved through rainfall
- ✓ **e.g.** Ageratum conyzoides, Amaranthus, Commelina benghalensis, Digera arvensis, Setaria glauca, Trianthema monogyna/ portulacastrum are rainy/wet season annuals.



Amaranthus

4. **Winter season annuals (Rabi annuals)**

- ✓ They grow **mainly during the winter/cold season** e.g. Chenopodium album, Melilotus spp., Phalaris minor, Avena fatua, Bromus pectinatus, Snowdenia polystachya are winter season annuals
- ✓ ******Simple annuals:** They are **incapable of growing back once they are cut off from the ground**. When chopped at **ground level**, **Parthenium, Lantana, and Pluchea** species appear to be perennials. Once more, they will grow from crown buds.



Phalaris minor

2. **Biennials -**

Biennial weeds typically **live longer than one but fewer than two seasons or years, during which they complete their life cycle**. In the first season or year, they form **rosettes and stay vegetative**; in the second season or year, **they produce flowers and set seeds**

Eg - Puncture vine, Chicory, Bull thistle



Chicory

3. Perennials:

Before they shrivel up or die, they grow for longer than two years. They bloom for the first time in their **second year of life**, after which they continue to grow from the same root system and bloom annually after that. The following categories apply to perennial weeds.



Lantana camara

✦ **Simple perennials:**

Although they are perennials, they primarily reproduce by seeds and, unless they are damaged or chopped, have no natural way of spreading vegetatively. For example, Sonchus arvensis, Oxalis latifolia, Lantana camara, Rumex spp.

✦ **Bulbous perennials:**

Although they can reproduce by seeds, they primarily reproduce by vegetative methods like stolons, rhizomes, tubers, bulbs, bulbils, corms, roots, stems, and leaves, which are very challenging to manage. e.g. Imperata cylindrica, Cynodon dactylon, Cyperus rotundus, Chromolaena odorata etc.



Cyperus Rotundus

✦ **Creeping perennials:**

They spread via rhizomes, stolons, spreading roots, and seeds, among other methods e.g.

✦ **Woody perennials:**

They produce woody biomass, stems, and branches, for example, by growing steadily and continuously over seasons or years and experiencing an annual growth increment by the compound interest law. Lantana camara, Prosopis juliflora.

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↓ **Corm perennials:**

Plants that **reproduce by corm and seeds**, such as *Phleum pratense* (Timothy), and have a modified shoot and fleshy stem.

3. Based on Morphology

Weeds can be **classified into two primary categories** based on their morphology and cotyledon characteristics: monocotyledonous and dicotyledonous weeds.

1. **Monocotyledonous weeds:**

weeds whose **seeds have a single cotyledon** and are incapable **of splitting in two**. They are divided into

- ✓ **Grasses:** All weeds belong to the **Poaceae family**, which is also known as **grasses** because of their long, narrow, spine-studded leaves.
Ex- *Phalaris minor*, *Cynodon dactylon*, *Commelina*, *Eichhornia crassipes* etc.



Cynodon Dactylon

- ✓ **Sedges:** This group **includes weeds** that belong to the Cyperaceae family.
The majority of the **leaves are basal and have modified stems that may or may not have tubers**. Sedges include **Cyperus** etc

2. **Dicotyledonous weeds:**

Dicotyledonous weeds have seeds **that can be divided into two halves and have two cotyledons**.

Ex *Trianthema portulacastrum*, *Amaranthus viridis*, *Chenopodium album*, *Melilotus indica*, *Striga asiatica*, *Spergula arvensis*, *Plantago lanceolata* etc.

3. Based on Nutrition habits/nature of competition

Weeds can be classified based on their nutritional habits or the nature of competition into two main categories:

1. Autotrophs/Nonparasitic weeds:

They produce **their food by themselves**. e.g. *Phalaris minor*, *Avena sativa*, *Commelina benghalensis* etc.

2. Heterotrophs/Parasitic weeds:

They are **dependent on crops or other sources of food** because **they** are unable to produce their food.

Depending on how **they parasitize roots and shoots**, they can be categorized once more in the following ways:

- ✓ **Complete Root parasitic weeds:** Root parasites have no other means of obtaining food and are entirely dependent on their host. **Ex - *Orobanch* spp**
- ✓ **Partial root parasitic weeds:** Partial root parasites are those weeds that initially depend upon the host roots for their food and living e.g. *Striga*
- ✓ **Complete stem parasitic weeds:** are those that take away food from the like *Cuscuta*. ***Cuscuta* is the only parasitic genus** of the autotrophic family *Convolvulaceae* now separated as the *Cuscutaceae* family.
- ✓ **Partial stem parasitic weeds:** Partial stem parasites are weeds that, after becoming green and chlorophyllous, rely on the host shoot or stem for nourishment. Examples of



Orobanch



Cuscuta

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such parasites are *Loranthus longiflorus* and *Cassytha filiformis*.

4. Based on Origin Weeds

Indigenous weeds:

A specific nation or continent is home to a vast number of native weeds that have been growing there for a very long time and are quite diverse.

For example, *Striga asiatic* is native to Asia while *Parthenium hysterophorus* is native to tropical North America.

Introduced or Exotic weeds or Alien weeds:

Weeds that started in one nation but eventually spread to another through dispersal are considered exotic or introduced weeds in that nation because they are not native to that nation.

Weeds like *Mikania micrantha* from Malaysia and *Sorghum halepense* from the USA were introduced in India.

6. Based on Association with crops

Season bound weeds:

These weeds only appear during a particular season of the year, regardless of the type of crop that is cultivated or grown. Season-specific weeds include *Fumaria* sp. and *Poa annua*, among others.

Crop-bound weeds:

These weeds are characterized by their partial or complete parasitism of a crop for food and survival.

For example, *Striga* sp. is restricted to sorghum crops, whereas *Loranthus longiflorus* is limited to tea plantations.

Crop-associated weeds:

Unlike weeds bound to crops, crop-associated weeds are not parasitic.

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There are specific reasons why they are connected to the crops.

Need for specific microclimate:

- ✓ Certain weeds **need a shady, cool, and moist** habitat to grow and survive more, so they congregate in areas where these conditions exist.
- ✓ For the **same reason that *Coronopus didymus*** is connected to lucerne, *Cichorium intybus* is linked to berseem.

Ready contamination with crops and weeds:

- ✓ Certain weeds, such as *Cichorium intybus* in berseem, *Avena ludiciana* (wild oat), *Convolvulus arvensis* (Hiran khuri) in wheat, *Phalaris minor* (canary grass), and *Oryza sativa* var. *fatua* (wild rice) in rice, mature their seeds **nearly at the same height and time as the corresponding crops**.
- ✓ These weeds' seeds resemble related crop seeds morphologically. As a result, **they can readily contaminate crop seeds during harvest** and are impossible to separate using any technique.

Weed mimicry with crops:

- ✓ Seed mimicry refers to the phenomenon of some seeds having characteristics **similar to those of a crop**, such as *Avena fatua* seeds resembling cultivated oat seeds, *Camelina sativa* seeds resembling flax seeds. Another name for them is "satellite weeds."
- ✓ "Certain weeds exhibit what's known as 'chronological mimicry,' whereby their height and maturity period resemble those of crop plants. *Phalaris minor*, for instance, reaches nearly the same maturity as wheat crops.

7. Based on Habitat/Situation of growing/Ecology

- ✓ **Terrestrial weeds:** Terrestrial weeds are those that are grown on land, such as those found in crop fields, fallow areas, and non-crop waste areas.
- ✓ **Xerophytic weeds:** Weeds growing in arid or semi-arid areas have very low water requirements and high-water use efficiency e.g. *Prosopis juliflora*, *Opuntia* spp.

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- ✓ **Mesophytic weeds:** Mesophytic weeds/plants are medium water requiring plants. e.g. *Cynodon dactylon*, *Cyperus rotundus* etc.

1. Cultivated/Arable crop weeds:

Arable crops are primarily those horticultural and agronomic plants that need to be sufficiently cultivated to germinate and continue growing.

↓ **Facultative/Apophytes weeds:**

"Facultative weeds" are weeds that grow mainly in wild communities but frequently invade cultivated areas and are closely associated with human activities, primarily crop cultivation. Examples of such weeds are *Euphorbia thymifolia* and *Anagallis arvensis*.

↓ **Obligate/Anthropophytes weeds:**

"Obligate weeds" are plants that are found mostly in cultivated or otherwise disturbed land and are never found in their natural state anywhere. Examples of these plants are *Lolium temulentum* and *Convolvulus arvensis*.

↓ **Orchard/Plantation weeds:**

"Plantation and orchard weeds" are weeds connected to plantation and orchard crops. For example, *Parthenium hysterophorus*, *Paspalum conjugatum*, and *Chromolaena odorata*.

↓ **Pasture/Grassland/Lawn weeds:**

"Lawn weeds" are weeds that are growing or existing in pastures, grasslands, or lawns. For example, *Melilotus* sp., *Poa annua*, and *Cynodon dactylon*.

2. Non-cropped/Wasteland/Roadside weeds:

Wasteland weeds are weeds that grow or are present on non-crop, waste, or fallow lands. Wasteland weeds can either be rudimentary or esoteric in their habits.

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Aquatic weeds:

Aquatic weeds are **weeds that grow in bodies of water such as lakes, ponds, drainage ditches, or main irrigation canals and lakes.**

They can be floating, emergent, or submerged, and they must complete at least a portion of their life cycle in water.

The following categories are typically used to classify aquatic weeds:

i) **Algae:** Algae are basic photosynthesizing plants that lack vascular tissues and stem, root, and leaf differentiation.

(a) Planktonic or Phytoplanktonic

(b) Filamentous algae

ii) **Vascular hydrophytes:** There are more than **100 known families of vascular hydrophytes**. In contrast to **mesophytes**, **these plants lack a strong stem** and well-developed conducting tissue.

They fall under the following categories:

1. Submersed aquatic weeds:

Submersed weeds are **those that remain below water surface**. Most of the vegetative growth of these weeds take place beneath the water surface.

They are further differentiated into:

✓ **Rooted submersed:** These weeds have roots, e.g.

Hydrilla verticillata (hydrilla) Valisneria spiralis (tape grass or eel weed), Potamogeton spp. (pond weed).

✓ **Suspended submersed:** These weeds have no roots, e.g. Ceratophyllum demersum (coon tail) and Utricularia spp. (bladderwort).



Hydrilla verticillata

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✓ Emerged aquatic weeds (Surface weeds):

In contrast to **floating aquatic weeds**, these plants can emerge or grow out of water with aerial stems and leaves at or above the water's surface, but their roots stay impregnated into the mud at the bottom of bodies of water
e.g. *Nelumbo nucifera*, *Typha*



Hydrilla verticillata

iii) Marginal aquatic weeds:

Some of them, like *Alternanthera axillaris* and *Fimbristylis miliacea*, grow close to the edges or **embankments of bodies of water or along shorelines** that are consistently moist and marshy with shallow water ponding that varies with the seasons.

iv) Floating aquatic weeds:

which **float freely, their roots submerged in water** and **their leaves above the surface**. Either individually or in rosettes, their leaves float on the water's surface. They have leaves and roots.

They are further differentiated into:

- ✓ Free floating weeds: *Eichornia*, *Pistia*, *Salvinia*.
- ✓ Rooted floating weeds: *Nymphaea*.



Eichornia



Nymphaea

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3. Based on Relative position/Presence

Weeds can be classified based on their Relative position/Presence into three main categories:

- ✓ **Absolute weeds:** Wild plants that are worthless as crops and of no use to farmers whatsoever, such as Rumex spp.
- ✓ **Relative weeds:** Despite not being weeds in the traditional sense, they are notable because of their irregular appearance or proximity to a specific crop, similar to mustard in a wheat field.
- ✓ **Rouge weeds:** plants of different varieties, such as rice grown between Rasi varieties and Jaya varieties, are deemed weeds when they are planted between varieties of primary importance.